

Algorithmic Curation and Moderation Effects on Political and Economic Discourse: A Cross-Platform Computational Study of UK Social Media

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1. Background and Motivation

Social-media platforms are now central to how citizens encounter and engage with political and economic information. Two algorithmic systems **curation** (ranking, recommendation) and **moderation** (labelling, down-ranking, removal) increasingly determine which narratives gain visibility and which are suppressed. While these mechanisms are justified as safeguards against misinformation and harm, they also influence the visibility and tone of democratic discourse (Gorwa 2020; Habibi et al. 2024).

Recent research highlights that algorithmic systems are not politically neutral. Huszár et al. (2022) found that Twitter's algorithm amplified mainstream right-leaning content more than left-leaning sources across most countries, though not the extremes, demonstrating systematic asymmetry in political amplification. Kelm (2023) and Metzler (2023) emphasised that these effects emerge from interactions between algorithmic design and social networks. Yet, there remains a lack of comprehensive, **cross-platform evidence** about how curation and moderation together shape discourse within specific national contexts like the United Kingdom, especially during key policy and electoral events. This question is increasingly relevant as Ofcom's remit expands under the **Online Safety Act (2023)** and as policymakers seek transparency in algorithmic decision-making.

Despite growing attention, the existing empirical record is fragmented. Studies tend to be **single-platform** (often Twitter/X), **descriptive**, and seldom address causal relationships between algorithmic interventions and public sentiment. Consequently, regulatory bodies and researchers lack clear metrics for assessing how algorithmic visibility and moderation decisions influence civic engagement and trust. This PhD proposes to fill that gap by combining **computational social science methods**, **causal inference**, and **large-scale social-data engineering** to study the democratic implications of algorithmic systems in the UK.

2. Research Aims and Novel Contribution

Aim

To investigate how algorithmic curation and content moderation influence the visibility, tone, and polarisation of political and economic discourse across major UK social-media platforms.

Research Significance and Novelty

This research will make four main contributions:

1. **Cross-platform focus:** By analysing multiple platforms (Twitter/X, Reddit, YouTube, Instagram and Facebook), this project transcends the limitations of single-platform analyses, producing generalisable insights into algorithmic governance.
2. **Causal measurement:** It will employ quasi-experimental designs (difference-in-differences, interrupted time-series) to isolate the causal impact of moderation and curation decisions on sentiment and engagement.
3. **Explainable NLP integration:** Using fine-tuned transformer models (BERT, RoBERTa) alongside explainability techniques (LIME, SHAP), the research will make algorithmic effects interpretable for scholars and regulators alike.
4. **Policy relevance:** It will generate actionable insights for UK institutions such as Ofcom and the Electoral Commission, aligning with current regulatory debates on algorithmic transparency.

By combining these methodological innovations, the project will deliver both theoretical advancement and practical frameworks for **algorithmic accountability** in digital political communication.

3. Research Questions

1. How do algorithmic curation mechanisms (ranking, recommendations) alter exposure and engagement with UK political and economic discourse?
2. How do moderation interventions (labelling, down-ranking, content removal) affect sentiment, participation, and discourse diversity?
3. How consistent are these effects across platforms with differing algorithmic architectures and content policies?

4. What linguistic or contextual features predict algorithmic amplification or moderation, and do these introduce systematic biases?
 5. Can we forecast sentiment and engagement dynamics surrounding key policy events and moderation interventions?
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4. Literature Context

Algorithmic Curation and Polarisation

Earlier studies on filter bubbles (Bakshy et al. 2015) raised concerns about selective exposure, but later work suggests complex interplays between algorithmic design and user behaviour. Huszár et al. (2022) provided large-scale quantitative evidence of political amplification biases in Twitter’s recommender system, while Kelm (2023) experimentally demonstrated that personalised feeds can increase affective polarisation. Lasser et al. (2025) proposed normative frameworks for designing civic-oriented recommender systems and highlighted the need for empirical evaluation of such designs, a challenge this PhD directly addresses.

Moderation and Democratic Quality

Automated moderation systems must balance harm reduction and expressive diversity. Habibi et al. (2024) demonstrate that content removal can inadvertently distort online discourse by shifting topic distributions and silencing certain perspectives, while Bouchaud and Ramaciotti (2025) find that crowd-sourced moderation systems, such as X’s Community Notes, perform unevenly across political contexts, particularly struggling to address highly polarising content. Cohen (2025) argues that algorithmic bias itself becomes a political topic in digital spaces. Casas (2024) highlights the geopolitical consequences of account suspensions, illustrating the power of moderation to shape participation and representation.

Cross-Platform Governance and Transparency

Comparative analyses (Metzler 2023; Jung 2024) underscore that algorithmic outcomes depend not only on design but also on civic context, Jung shows that algorithmic exposure can translate into offline civic engagement, linking digital design choices to real-world participation. Casas, Dagher, and O’Loughlin (2025) emphasise the importance of academic access to platform data for advancing online safety and accountability research. However, the UK lacks a unified computational framework linking platform-level moderation events to shifts in sentiment and engagement. This project aims to fill that gap by uniting cross-platform analytics with policy-relevant causal modelling.

5. Methodology

5.1 Research Design

This project employs a **computational mixed-methods approach** integrating:

- **Large-scale observational analysis** of historical and real-time social-media data.
- **Quasi-experiments** (e.g., difference-in-differences, interrupted time-series) around major moderation or curation events.
- **Audit studies** comparing algorithmic versus chronological content feeds.

5.2 Data Collection

This research will employ a multi-source, triangulated data strategy to ensure both coverage and methodological robustness. Data will be collected through a combination of **official APIs**, **licensed social listening platforms**, and **ethical web data providers**, balancing representativeness, reproducibility, and compliance with platform regulations. By combining these sources, the study will enable both **historical reconstruction** (via archives and scraping frameworks) and **real-time monitoring** of content flows, ensuring temporal depth and reliability across case studies.

- **Platforms:** Twitter/X, Reddit, YouTube, Instagram, and Facebook through social listening tools like Brandwatch or third party tools like Apify, Bright Data, Zyte etc.
- **Topics:** UK political and economic issues - elections, budgets, taxation, welfare, inflation, energy.
- **Signals:** Labelling, removal, down-ranking, visibility limits, and engagement changes.
- **Ethics:** Public data only, fully anonymised, GDPR-compliant, and within platform terms of service.

5.3 Analytical Framework

Data processing will follow a scalable **AWS-based architecture** (S3 → Glue → RDS → SageMaker) orchestrated with Airflow/Step Function, ensuring transparency and reproducibility. Analyses will integrate the following components:

- **Detection:** Pattern recognition of moderation or curation events using metadata (e.g., label_added, policy_violation).
- **Sentiment and Stance Analysis:** Fine-tuned BERT and RoBERTa models calibrated with lexicons (VADER, NRC) for cross-validation.
- **Topic Modelling:** BERTopic and zero-shot classification to capture issue framing.

- **Explainability:** LIME and SHAP to surface linguistic cues behind algorithmic interventions.
- **Causal Inference:** Difference-in-differences, propensity-score matching, and synthetic control to estimate moderation effects.
- **Forecasting:** Prophet and LSTM networks to simulate sentiment trajectories.

5.4 Validation and Robustness

Manual annotation of 1,000 randomly sampled posts (Cohen's $\kappa \geq 0.75$) will ensure model validity. Bootstrapping and McNemar tests will assess classifier reliability. Sensitivity analyses will test the robustness of causal inferences across alternative model specifications.

6. Proposed Chapter Structure (Three-Paper Model)

Paper 1: Algorithmic Curation and Political Exposure

This paper will map the visibility and amplification patterns of UK political content across multiple platforms. By contrasting algorithmic feeds with chronological timelines, it will quantify how ranking and recommendation systems alter user exposure and engagement. Building on the frameworks of Huszár et al. (2022) and Metzler (2023), it will estimate whether algorithmic curation amplifies particular ideological or emotional frames.

Paper 2: Moderation and Discourse Quality

This paper will assess the causal effects of moderation interventions (labelling, down-ranking, content removal) on sentiment, toxicity, and participation. Using quasi-experimental designs (difference-in-differences and interrupted time-series), it will identify whether moderation dampens polarisation or suppresses legitimate civic expression, contributing empirical depth to Habibi et al. (2024) and Bouchaud and Ramaciotti (2025).

Paper 3: Economic Policy Discourse Across Platforms

Extending the researcher's MSc work, this paper will examine cross-platform discourse on UK economic policies (e.g., budgets, inflation, welfare). It will combine algorithmic intervention data with transformer-based sentiment models to explore how moderation and curation jointly influence public understanding of economic governance. This final study will integrate predictive modelling (Prophet/LSTM) to forecast discourse responses to policy announcements.

7. Expected Contributions

Academic Contributions

1. The first UK-focused, cross-platform quantitative study linking algorithmic curation and moderation to political discourse.
2. A methodological contribution uniting causal inference, explainable NLP, and computational data engineering.
3. Extension of existing frameworks (Casas 2024; Casas et al. 2025) by empirically connecting moderation signals to sentiment dynamics.

Policy and Practical Contributions

1. Evidence-based insights for Ofcom and the Electoral Commission on transparency and algorithmic risk assessment.
2. A research-grade monitoring dashboard visualising real-time sentiment and moderation indicators along with robust data pipeline for reproducibility.
3. Open-source datasets and codebases enabling replication and regulatory benchmarking.

8. Ethics and Risk Management

All data will be sourced from public APIs, anonymised, and handled in compliance with GDPR and institutional ethics protocols. Sensitive content will be filtered, and researcher wellbeing will be prioritised. Bias detection and fairness audits will assess model performance across dialects and topics. Multiple identification strategies (quasi-experimental triangulation) will mitigate causal overreach.

9. Timeline (3.5 Years)

Year	Key Activities
1	Literature review, ethics approval, data infrastructure setup, pilot analysis (one platform).

2	Curation and moderation analyses; Paper 1 submission; causal modelling design for Paper 2.
3	Cross-platform integration, forecasting models; Paper 2 and 3 drafts; conference presentations.
3.5	Thesis synthesis, policy briefs, dissemination, and viva preparation.

10. Conclusion

This PhD will provide systematic, causal, and interpretable evidence on how algorithmic moderation and curation shape democratic discourse in the UK. It will bridge computational social science with regulatory policymaking, advancing both academic theory and practical governance. Ultimately, the research aims to inform transparent, accountable, and citizen-oriented algorithmic systems laying a foundation for evidence-based oversight in the digital public sphere.

References

1. Bakshy, E., Messing, S. and Adamic, L.A. (2015) 'Exposure to ideologically diverse news and opinion on Facebook', *Science*, 348(6239), pp. 1130-1132.
2. Bouchaud, P. and Ramaciotti, P. (2025) 'Algorithmic resolution of crowd-sourced moderation on X in polarised settings across countries', *arXiv preprint arXiv:2506.15168*.
3. Casas, A. (2024) 'The geopolitics of deplatforming: A study of suspensions of politically interested Iranian accounts on Twitter', *Political Communication*, 41(3), pp. 321-344.
4. Casas, A., Dagher, L. and O'Loughlin, M. (2025) 'Academic access to social media data for the study of political online safety', Royal Holloway University of London Working Paper.
5. Cohen, M. (2025) 'Hijacking algorithmic bias: Analysing the political discourse around ChatGPT on social media', *Information, Communication & Society*, 28(5), pp. 712-730.
6. Gorwa, R. (2020) 'Algorithmic content moderation: Technical and political challenges', *Big Data & Society*, 7(1), pp. 1-15.
7. Habibi, M., Hovy, D. and Schwarz, C. (2024) 'The content moderator's dilemma: Removal of toxic content and distortions to online discourse', *arXiv preprint arXiv:2412.16114*.

8. Huszár, F. et al. (2022) 'Algorithmic amplification of politics on Twitter', *PNAS*, 119(1), e2025334119.
 9. Jung, H. (2024) "How social media algorithms shape offline civic engagement," *Computers in Human Behavior Reports*, 12, 100211.
 10. Kelm, O. (2023) "How algorithmically curated online environments influence political polarisation," *Computers in Human Behavior Reports*, **11**, 100343.
 11. Lasser, J. et al. (2025) "Designing social media content recommendation algorithms for civic discourse," *Annals of the New York Academy of Sciences*, 1522(1), pp. 27–41.
 12. Metzler, H. (2023) 'Social drivers and algorithmic mechanisms on digital media', *Frontiers in Communication*, 8, 1242173.
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Word count: ~1,881 words